「Def. Let S be a regular surface and T_pS be the tangent plane at $p\in S$.
A quadratic form

$$I_p : T_pS \to \mathbb{R} : w \mapsto \langle w, w\rangle_p \geq 0$$

is called the first fundamental form of S at p.

「Computation of the fundamental form.
Given $p\in S$, let $X : U \to S$ be a parametrization of S containing p.
Let $w \in T_pS$ be given. Take a curve

$$\alpha : (-\varepsilon, \varepsilon) \to X[U] : t \mapsto X(u(t), v(t))$$ such that $\alpha(0) = p$ and $\alpha'(0) = w$.

Now,

$$\begin{aligned} I_p(w) = I_p(\alpha'(0)) &= \langle \alpha'(0), \alpha'(0)\rangle_p \\ &= \langle X_u u' + X_v v', X_u u' + X_v v'\rangle \\ &= \langle X_u, X_u\rangle\cdot(u')^2 + 2\cdot\langle X_u, X_v\rangle\cdot u'v' + \langle X_v, X_v\rangle\cdot(v')^2. \quad (t=0). \end{aligned}$$

In this sence, define

$$E(u_0, v_0) = \langle X_u, X_u\rangle_p, \quad F(u_0, v_0) = \langle X_u, X_v\rangle_p, \quad G(u_0, v_0) = \langle X_v, X_v\rangle_p.$$

「Def. A length of parametrized curve $\alpha : I \to S$ is

$$S(t) = \int_0^t |\alpha'(r)|\, dr = \int_0^t \sqrt{I(\alpha'(r))}\, dr = \int_0^t \sqrt{E\cdot(u')^2 + 2F\cdot u'v' + G(v')^2}\, dr$$

「Def. Let $R \subset S$ be a bounded set contained a image of $X : U \to S$,
A area of R is:

$$A(R) = \iint_{X^{-1}[R]} |X_u \times X_v|\, du\, dv.$$

「Examples,

Let $S=\{(x,y,z) \mid x^2+y^2=1\}$ be the Sylinder.

Consider a Parametrization of S,

$$X : U \to \mathbb{R}^3 : (u,v) \mapsto (\cos u, \sin u, v)$$

where $U=\{(u,v)\in\mathbb{R}^2 \mid 0<u<2\pi, -\infty<v<\infty\}$.

Simply, $X_u=(-\sin u, \cos u, 0)$ and $X_v=(0,0,1)$.

So, $E=1$, $F=0$, $G=1$.

「Helicoid.

Let $S=\{(r\cdot\cos\theta, r\cdot\sin\theta, a$

Exercises. Sec 2.5.

1. Evaluate the first-fundamental form of the following:

a). $X(u,v) = (a \cdot \sin u \cdot \cos v,\ b \cdot \sin u \cdot \sin v,\ c \cdot \cos u)$

b). $X(u,v) = (a \cdot u \cos u,\ b \cdot u \sin v,\ u^2)$

Proof.

a), $X_u = (a \cos u \cdot \cos v,\ b \cos u \cdot \sin v,\ -c \cdot \sin u)$,

$X_v = (-a \cdot \sin u \cdot \sin v,\ b \cdot \sin u \cdot \cos v,\ 0)$.

So, $E = \langle X_u, X_u \rangle = a^2 \cdot \cos^2 u \cdot \cos^2 v + b^2 \cos^2 u \cdot \sin^2 v + c^2 \cdot \sin^2 u$

$= (a^2 \cos^2 v + b^2 \sin^2 v) \cos^2 u + c^2 \cdot \sin^2 u$

$F = \langle X_u, X_v \rangle = -a^2 \cdot \cos u \cdot \sin u \cdot \cos v \cdot \sin v + b^2 \cdot \cos u \cdot \sin u \cdot \cos v \cdot \sin v$

$= (-a^2 + b^2) \cdot \cos u \cdot \sin u \cdot \cos v \cdot \sin v$.

$G = \langle X_v, X_v \rangle = a^2 \cdot \sin^2 u \cdot \sin^2 v + b^2 \cdot \sin^2 v \cdot \cos^2 v$.